

GUJARAT UNIVERSITY
B.E. Sem. – III (IT) Examination
Data Structures & Algorithms

Wednesday, 29th November, 2006]

[Time: 10-30 a.m. to 1-30 p.m.

[Max.Marks : 100

- Instructions :** (1) Attempt all questions.
 (2) Answer to the two sections must be written in separate answer-books.
 (3) Figures to the right indicate full marks.
 (4) Assume suitable data, if required.

SECTION I

- 1 (a) Explain time & space complexity of an algorithm with example 6
 (b) What will be the postfix notation of $A+(B*C-(D/E-F)*G)*H$ 6
 (c) Write an algorithm to convert an infix expression into postfix expression 6

- 2 (a) Write a function to insert & delete an item from the circular queue 6
 (b) Explain the classification of data structure 4
 (c) Write a program to implement doubly linked list 6

OR

- 2 (a) Write an algorithm to insert an item in last in first out linked list 6
 (b) Write a program to search a given element in list using binary search method 8
 (c) Define priority queue 2

- 3 Answers the following (any two) 16
 (a) Write a program to implement stack as a linked list
 (b) Explain various multiple key access file organization
 (c) Explain merge sort with example

SECTION II

- 4 (a) Define the following terms: 6
 (1) Thread
 (2) Adjacency matrix
 (3) Hash function

- (b) Define B-tree. Explain with an example how will you insert & delete nodes on a B-tree 6
 (c) Write & explain non-recursive algorithm for a pre-order traversal of a binary tree 6

- 5 (a) Explain radix sort with example 6
 (b) Sort the following array using quick sort 6
 12 48 37 64 71 18 33 55 47 19 21
 (c) Discuss the sorting time for various methods of sorting 4

OR

- 5 (a) What is meant by Collision in a hashing scheme? What are ways of resolving them ? 6
 (b) What is hashing? List out important features of a good hashing algorithm 6
 (c) Explain indexed sequential search method 4

- 6 Answers the following (any two) 16
 (a) Explain kruskal's algorithm for minimum cost of spanning tree
 (b) Write an algorithm to construct a threaded binary tree
 (c) Explain BFS & DFS with example